Fossils reviewed



Palaeontology is the science of fossils (*paleo* ancient). A *fossil* can be defined as any trace of a plant or animal that has been preserved in the Earth's crust, most commonly in rock or sand. Fossils can be formed as imprints, skeletal remains or by an organism being petrified. Often only the hard parts of organisms (living things), for example bones, shell or bark tend to be preserved.

A fossil can be made from a porous (contains lots of small holes) object such as a bone. Under the right conditions water containing minerals can fill the porous parts. Over a long period of time this can make the object resemble stone and preserves it. In a similar way, a piece of wood can have its molecules replaced very slowly by minerals, making it petrified.

1 *Petrified* literally means 'turned to stone'. Give another meaning of *petrified*.

In rare cases, such as examples of the Woolly Mammoth, entire bodies (including stomach contents!) have been found preserved in Europe.

2 Which modern (but less hairy) mammal does the Woolly Mammoth resemble?

3 Suggest why *palaeontology* and *geology* are closely linked.

4 Suggest why, in places like Siberia, entire bodies have been preserved.

Fossils form when materials from the Earth's crust (sediment) cover the remains of an organism, either under water or on land. As greater amounts of deposit accumulate, the remains are compressed by the rock. Imprints (perhaps a footprint in soft soil) can also be filled in and covered up over time.

Various scientific tests can be performed to determine the ages of fossils, thus enabling us to develop theories as to how life on Earth evolved. Not only does finding a certain species in a given area suggest what lived there (and when), but it can also tell us about environmental conditions. For example, fossilised shells have been found in deserts suggesting that water once covered the land.

5 Many insect remains have been preserved in amber, a hardened, golden-brown tree resin. Suggest why these trapped insects did not decay.

Sometimes only part of an organism is fossilised when it dies.

6 What parts of plants are most likely to be found fossilised? Explain your answer.

Imprints, such as tracks and footprints, can form fossils. One example might be a dinosaur walking on a muddy riverbank leaving its footprints. The mud dries out and becomes covered in sand and other sediments that eventually form hard rock. Millions of years later the original footprints become uncovered. The size and distance between these footprints can give information about the dinosaur.

7 Explain how this information might help a palaeontologist work out the physical dimensions (weight, height etc) of a dinosaur.

Research

Fossils are classified into four groups. Investigate the following types then write short descriptions of how they are formed.

8 Casts and moulds **9** Indexed remains **10** Replaced remains **11** Impressions

Remember! If you find or collect fossils, be sure to record the location and date the fossil was found. Your fossil might one day help scientists to answer important questions about life in the past.

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